Longitudinal IMU-based Analysis of Biomechanical Adaptations during Robotic Gait Rehabilitation Therapy after Stroke

Irene Bono-Gutiérrez [1], Camila A. Castellanos Guarnizo[1], Agnese Cherubini[2], Natacha León[2], Jesús Tornero[2], Juan C. Moreno[1], Clara B. Sanz-Morère[1]

[1] BioRobotics Group, Center for Automation and Robotics, Spanish Research Council (CSIC), Madrid, Spain. jc.moreno@csic.es

[2] Advanced Neurorehabilitation Unit, Hospital Universitario Los Madroños, Madrid, Spain.

Introduction

Among non-communicable disorders, stroke remains the second leading cause of death and the third leading cause of disability worldwide [1]. Over the past 30 years, its incidence has increased by 70% [2], with only 60% of survivors regaining independent walking after conventional rehabilitation [3], with persistent compensatory gait patterns [4]. Robot-Assisted Gait Training offers safer, more repetitive, and less physically demanding rehabilitation for physiotherapists [5], but evidence of its superiority over conventional therapy remains inconclusive [6]. This may be partly due to limitations of clinical scales, which may overlook subtle changes in gait performance [6]. Objective biomechanical measures, including spatiotemporal and joint kinematic parameters, can complement clinical assessments and provide a more detailed characterization of gait recovery. This study aims to analyze the weekly evolution of these parameters in a stroke patient undergoing robotic gait rehabilitation.

Methods

The participant underwent 8 weeks of Walkbot rehabilitation. Functional improvements were assessed with the Functional Independence Measure (FIM) and the Trunk Control Test (TCT) before, in the middle, at the end of therapy, and after 2-months (FU). Additionally, once a week, the participant completed two 10-meter overground walking trials while wearing Xsens inertial measurement units. Kinematics were extracted from the software MVN Analyze, data were standardized to the EUROBENCH format, segmented based on gait events (heel strike and toe off), normalized, and aggregated. Range of motion (ROM) and gait parameters were extracted. Shapiro–Wilk, Kruskal–Wallis and Mann–Whitney tests were used to examine normality, differences across the weekly assessments and between weeks 1 and 9, respectively, for double-support (DS) and stride time (ST).

Results

The clinical scales indicated improvements throughout the rehabilitation period, with FIM and TCT increasing by 16% and 39%, respectively. These gains remained at the FU session, with a further 3% improvement observed in FIM. From week 1 to week 9, median ROM increased by 6.15° (22%), while ST and DS decreased by 0.15 s (12%) and 0.12 s (40%), respectively, and cadence increased by 14.73 steps/min (15%). Figure 1 illustrates the longitudinal evolution of biomechanical outcomes, with significant differences in ST and DS ($p < 0.05$), while also showing overall trends but with some intra- and inter-session variability. The latter was more evident in week 6 for ST and cadence, and in weeks 2, 4, and 6 for knee ROM, reflecting the patient's performance in response to the varying demands of rehabilitation.

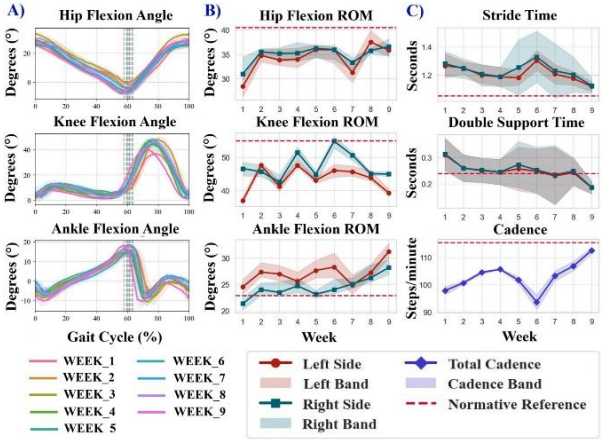


Figure 1: Biomechanical results obtained in every session (week 1 to 9). A) Joint angles from the affected limb (left side). B) Range of motion. C) Temporal parameters.

Conclusions

Weekly assessments of a post-stroke patient during Walkbot-assisted rehabilitation captured the evolution of their gait patterns, revealing quantitative changes in joint kinematics and temporal performance that may not be captured by conventional clinical scales alone. These results support IMU-based biomechanical monitoring as a low-cost approach for characterizing gait and its potential recovery during rehabilitation. Further analysis of the available data from additional participants will help assess the consistency of these findings and their causal relationship with the robotic rehabilitation.

References

1. M.Katan et al, *Semin Neurol*, 38:208–211, 2018.
2. eClinicalMedicine, eClinicalMedicine, vol 59, 2023.
3. S. Murphy et al, *Medicine* (Abingdon), 48:561–566, 2020.
4. J. Veerbeek et al, *Neurorehabilit Neural Repair*, 25:268–274, 2011.
5. P. de Andrade et al., *Physiother Res Int*, vol. 31, 2026.
6. L.Wu et al, *Front Neurol*, vol. 14, 2023.